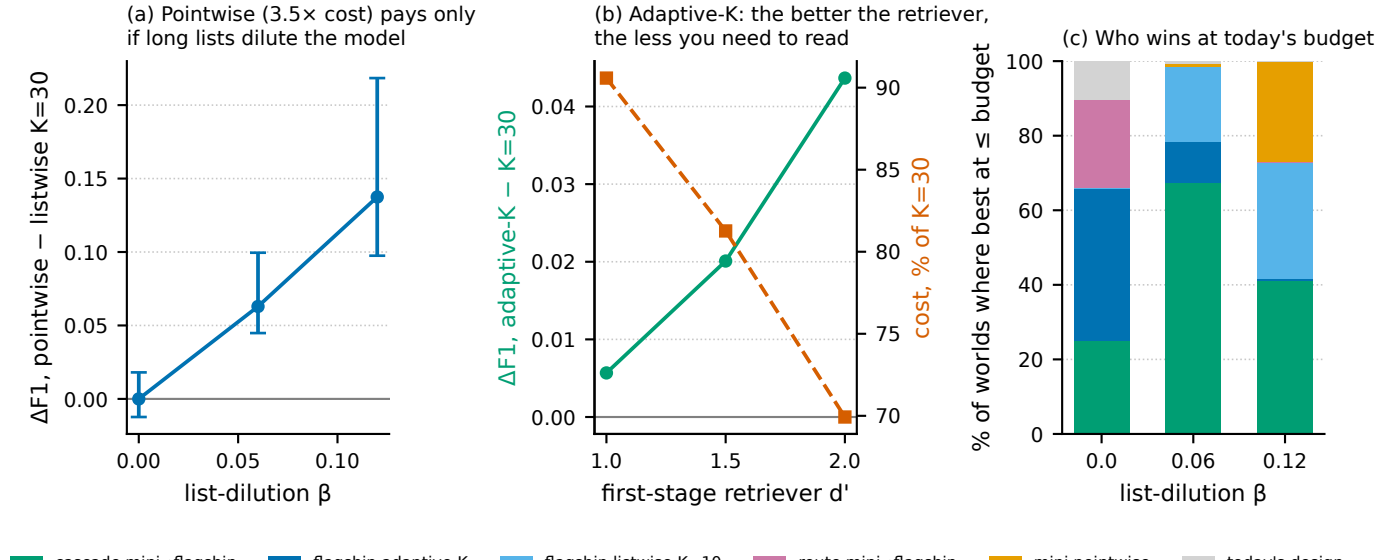
(a) Pointwise (3.5× cost) pays only if long lists dilute the model
ΔF1, pointwise − listwise K=30
0.20
0.15
0.10
0.05
0.00
0.00
0.05
0.10
list-dilution β
(b) Adaptive-K: the better the retriever, the less you need to read
ΔF1, adaptive-K − K=30
0.04
0.03
0.02
0.01
0.00
cost, % of K=30
90
85
80
75
70
1.0
1.5
2.0
first-stage retriever d'
(c) Who wins at today's budget
% of worlds where best at ≤ budget
100
80
60
40
20
0
0.0
0.06
0.12
list-dilution β
cascade mini→flagship
flagship adaptive-K
flagship listwise K=10
route mini→flagship
mini pointwise
today's design